The endpoint equality paths force an opposite head

Human Compression project: L044 continuation

Retain the complete critical-head hypotheses and acquired proofs, including the explicit smaller-degree alternating induction. In particular $n = 2r$, the head is supported on D, and

$$R = G_{\geq 3} = \mathrm{Alt}(\Omega \setminus (D \setminus \{6\})).$$

The actual prefixes $P_i = G_{\leq i}$ have their proved pure alternating kernels and exact intersection formula with R. Eligibility always concerns full ambient omissions.

Lemma 1 (All later marks fix a saturated prefix support)**.** *Suppose $g \geq 5$, $P = P_{g-1}$ has exactly two nontrivial orbits $K = C_{g-1}$ and E, and $|K| + |E| = 2g + 1$. Suppose every label $3, \ldots, g-1$ is paired. Then every ρ_k with $k \geq g + 1$ fixes $K \cup E$ pointwise.*

Proof. Such a mark $z = \rho_k$ centralizes P, fixes K pointwise by its pure natural alternating kernel, and preserves the only other nontrivial P-orbit E. If its restriction f to E is nonidentity, centralization and transitivity make f a free involution. Write $|E| = 2m$.

Set $I = \{3, \ldots, g-1\}$ and $H = G_I$. Its two nontrivial orbits are $T = K \setminus (D \setminus \{6\})$ and E, with $|T| = |K| - 5$, by the exact prefix intersection formula. Let $L \subseteq I$ be the labels having an ambient-eligible edge in E. Such an edge cannot be a pair of f, since $z \in G_i$ for every $i \in I$. Select one such edge per label in L and project to the m pairs of f. The selected graph is a forest: an avoiding path around a cycle lifts to E because H commutes with f, with $z|_E = f$ supplying any switches within a pair. It would give an actual avoiding path for an eligible edge. Thus $|L| \leq m-1$. Every eligible edge for a label in $I \setminus L$ lies in T. The acquired selected-localization theorem gives $2|I \setminus L| \leq |T| - 1 = |K| - 6$. Consequently

$$g - 3 \leq \frac{|E|}{2} - 1 + \frac{|K| - 6}{2} = \frac{2g - 7}{2} < g - 3,$$

a contradiction. Hence $f = 1$, proving the lemma. This is the exterior-deficit argument for an arbitrary later mark; no crossing or parity assumption on $z|_E$ is needed. □

Theorem 2 (The first later endpoint cut)**.** *Suppose $B = G_{\geq 4}$ is intransitive on $\Omega \setminus D$, and $r - 1$ is the least label at least five having a unique eligible edge. Suppose its first shared label is $g \geq 5$, and $P = P_{g-1}$ is intransitive on its support. Then $g = r - 3$, and the actual last three generators, in reversed order, form the pure degree-six A_5 marking of source Table 2(2). In particular the endpoint branch reduces to an actual opposite head, rather than a projected isomorphism.*

Proof. The accepted early-cut calculation and the present least-cut hypothesis make every label $3, \ldots, r-2$ paired. Write $C = C_{r-2}$ for the head-side omission orbit and A for the other orbit. All indices below g act on C alone: none is shared, and a change to a disjoint supported side would disconnect the noncommutation diagram. Thus P fixes A. Put $Y = \mathrm{supp}(P)$ and $h = r-1-g \geq 1$.

Apply the complete endpoint support-deficit lemma after full reversal. Its crossing is now index zero, its last shared index is h, and its remaining tail is exactly P. It gives $|\mathrm{Fix}(P)| \geq 2h + 1$, hence $|Y| \leq 2g + 1$. This is the endpoint theorem and its endpoint path equality, not an interior equality interface. If $p \geq 2$ counts the nontrivial P-orbits, then $H = P \cap R = G_{3,\ldots,g-1}$ has support $|Y| - 5$ and p nontrivial orbits. The acquired paired interval bound gives

$$2g - 6 \leq |Y| - 5 - p \leq 2g - 4 - p.$$

Therefore $p = 2$, $|Y| = 2g+1$, and P has head orbit K and one exterior orbit E. Lemma 1 applies.

Write $s = \rho_{r-1}$ and $u_i = \rho_{r-1-i}$ for $1 \le i \le h$. The exact endpoint equality gives disjoint paths

$$a_0 -_{u_1} a_1 -_{u_2} \cdots -_{u_h} a_h, \qquad b_0 -_{u_1} b_1 -_{u_2} \cdots -_{u_h} b_h,$$

where s crosses a_0, b_0,

$$A = \{a_0, \ldots, a_h\}, \quad \mathrm{Fix}(P) = A \cup \{b_0, \ldots, b_{h-1}\}, \quad b_h \in Y.$$

These statements describe the exact path vertices and fixed set; no unproved classification of all graph edges is inserted.

By Lemma 1, s fixes Y pointwise. If $h \ge 2$, start from b_h and commute s past $u_h, u_{h-1}, \ldots, u_2$ to see that it fixes $b_{h-1}, \ldots, b_1$ as well. Every one of these commutations is at distance at least two in the original marking.

For $h \ge 3$, the acquired exhausted anchored-path lemma, applied to $u_1, \ldots, u_h$ on their $h+1$ points A, gives the full natural symmetric action

$$\langle u_2, \ldots, u_h \rangle|_A = \mathrm{Sym}(A \setminus \{a_0\}).$$

Its hypotheses hold because the endpoint anchor a_0 is fixed by $u_2, \ldots, u_h$, the displayed path makes the action transitive, and the last mark is nonidentity there. The unique crossing makes s preserve $A \setminus \{a_0\}$; it centralizes the displayed natural S_h, whose centralizer is trivial for $h \ge 3$. Thus it fixes that set pointwise. We have accounted for every point: $s = (a_0\ b_0)$, contrary to evenness. For $h = 1$, the same conclusion follows because $A \setminus \{a_0\}$ is a singleton and Y is fixed.

It remains that $h = 2$, so $g = r - 3$. Here evenness of s, its fixed set $Y \cup \{b_1\}$ and its unique crossing give

$$s = (a_0\ b_0)(a_1\ a_2).$$

The mark $u_1 = \rho_{g+1}$ also fixes Y by Lemma 1. Its displayed path edges and the involution condition on the remaining single point of A give

$$u_1 = (a_0\ a_1)(b_0\ b_1), \qquad u_2 = (a_1\ a_2)(b_1\ b_2)\delta,$$

where $u_2 = \rho_g$ fixes b_0 by the unique crossing and far commutation, and δ is an even involution supported on $Y \setminus \{b_2\}$. These are exactly the accepted six-point Table 2(2) core permutations, in reversed order, with a possible disjoint exterior factor on the third mark. Their precise degree-six A_5 identification is already paid.

The prefix P_g is transitive on $Y \cup \{b_1\}$. Indeed it preserves that set and fixes b_0, while adjoining u_1 merely connects b_0 to b_1 on C; if any other P_g-orbit in that set remained, P_{r-2} would not be transitive on C. Its pure alternating kernel therefore contains δ.

If $\delta \ne 1$, the subgroup $Q = \langle s, u_1, u_2 \rangle$ projects onto the displayed perfect core A_5 and onto $\langle \delta \rangle$. If it did not contain the pure exterior δ, projection to the core would be injective and give a nontrivial homomorphism $A_5 \longrightarrow C_2$. This is impossible. Thus $\delta \in Q$. Actual IP then gives

$$\delta \in Q \cap P_g = \langle u_2 \rangle.$$

But δ fixes b_1 whereas u_2 moves it. Hence $\delta = 1$, a contradiction. The exterior factor is therefore trivial, and the actual opposite head is exactly as claimed. □

Attribution and boundary. The endpoint paths (source Proposition 5.20), exhausted symmetric path and exact perfect core are acquired proofs. The new step fixes the old support under every later mark. All predecessors remain paid, including the root's parity-fibre and global-middle contributions. The intransitive B and prefix hypotheses remain explicit. Opposite-head exclusion, the unshared singleton endpoint and complete L044 recovery remain separate. No new computation, formal theorem, whole-original or question credit is claimed.